

CampusEats

Service Contracts, placeOrder Specification & Validation

CS 543 · Web Services

CampusEats — Service Contracts

1. Accounts Service

Operation	Input	Output	Errors
getAddress	studentId, addressId	Delivery address	Address not found
getProfile	studentId	User profile	User not found

2. Catalogue Service

Operation	Input	Output	Errors
listRestaurants	Area or filter	List of restaurants	No restaurants found
getMenu	restaurantId	Menu items with prices	Restaurant not found
checkItem	itemId	Availability and current price	Item not found

3. Orders Service

Operation	Input	Output	Errors
addToCart	studentId, itemId, quantity	Updated cart	Item unavailable, invalid quantity
placeOrder	studentId, items, deliveryAddressId, paymentMethodId	Order reference, status, total, estimated time	Empty cart, invalid address, item unavailable, payment declined
getOrder	orderId	Order details and status	Order not found
cancelOrder	orderId	Cancellation status	Order not found, too late to cancel

4. Payments Service

Operation	Input	Output	Errors
charge	Order reference, amount, payment method	Payment receipt and status	Payment declined, invalid payment method

Operation	Input	Output	Errors
refund	Payment reference, amount	Refund status	Payment not found, refund rejected

5. Delivery Service

Operation	Input	Output	Errors
assignRider	Order reference, delivery location	Rider assignment and estimated time	No rider available, invalid order
getTracking	Order reference	Delivery status, location, estimated time	Delivery not found

6. Notifications Service

Operation	Input	Output	Errors
send	Recipient, message, notification type	Notification status	Invalid recipient, delivery failed

Central Operation — placeOrder

Purpose

placeOrder allows a student to submit selected food items for delivery. The Orders Service coordinates validation, catalogue checking, payment, rider assignment, and notification.

Input

Field	Description
studentId	Identifies the student placing the order
items	List of food items and requested quantities
deliveryAddressId	Address where the order should be delivered
paymentMethodId	Payment method selected by the student

Item Input

Each item contains:

Field	Description
itemId	Food item being ordered
quantity	Number of units requested

Example:

```
placeOrder(  
  studentId,  
  items: [  
    { itemId, quantity },  
    { itemId, quantity }  
  ],  
  deliveryAddressId,  
  paymentMethodId  
)
```

Service Validation — Five Properties

Each CampusEats service is evaluated against the five properties of a good web service.

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled
Accounts	✓	✓	✓	✓	✓
Catalogue	✓	✓	✓	✓	✓
Orders	✓	✓	✓	✓	✓
Payments	✓	✓	✓	✓	✓
Delivery	✓	✓	✓	✓	✓
Notifications	✓	✓	✓	✓	✓

Validation Details

Accounts Service

Property	Result	Reason
Reachable	✓	Other services can call its published operations over the network.
Self-contained	✓	It manages user profiles and delivery addresses as its own capability.
Has Contract	✓	Operations such as getAddress() and getProfile() define its interface.
Independent	✓	Its internal implementation can change without changing callers.
Loosely Coupled	✓	Other services request account information through contracts rather than accessing its data directly.

Catalogue Service

Property	Result	Reason
Reachable	✓	Other services can call catalogue operations over the network.
Self-contained	✓	It manages restaurants, menus, prices, and item availability.
Has Contract	✓	listRestaurants(), getMenu(), and checkItem() define its contract.

Property	Result	Reason
Independent	✓	Catalogue implementation can change without changing Orders.
Loosely Coupled	✓	Orders uses checkItem() instead of accessing catalogue data directly.

Orders Service

Property	Result	Reason
Reachable	✓	Students and other services can access its published operations.
Self-contained	✓	It manages carts, orders, and order status.
Has Contract	✓	addToCart(), placeOrder(), getOrder(), and cancelOrder() define its contract.
Independent	✓	Its internal order implementation can change without exposing its data to other services.
Loosely Coupled	✓	It communicates with Accounts, Catalogue, Payments, Delivery, and Notifications through contracts.

Payments Service

Property	Result	Reason
Reachable	✓	Orders can call payment operations over the network.
Self-contained	✓	It performs payment and refund responsibilities itself.
Has Contract	✓	charge() and refund() define its public contract.
Independent	✓	The payment provider can be changed internally without changing Orders.
Loosely Coupled	✓	Orders receives payment results through charge() instead of accessing payment data.

Delivery Service

Property	Result	Reason
Reachable	✓	Orders can call delivery operations over the network.

Property	Result	Reason
Self-contained	✓	It manages riders, assignments, and delivery tracking.
Has Contract	✓	assignRider() and getTracking() define its public operations.
Independent	✓	Rider assignment and tracking implementation can change internally.
Loosely Coupled	✓	Orders requests rider assignment through a contract instead of accessing rider data.

Notifications Service

Property	Result	Reason
Reachable	✓	Other services can request notifications over the network.
Self-contained	✓	It performs the notification delivery responsibility.
Has Contract	✓	send() defines the operation available to other services.
Independent	✓	Its notification provider or implementation can change without changing callers.
Loosely Coupled	✓	Services call send() rather than accessing the notification log directly.

Overall Result

All six CampusEats services satisfy the five required service properties.

The design maintains independent data ownership and uses service contracts for communication. No service directly accesses another service's internal data.